

Abstract No.

How dendritic cell maturation instructs leptomeningeal anti-tumor immunity

Fabrice A. Nkengeh¹, Xinran Tong², Adrienne Boire², and Jan Remsik^{1,}*

¹VIB Center for Cancer Biology

²Memorial Sloan Kettering Cancer Center

*Presenting author

Metastasis to the cerebrospinal fluid-filled leptomeninges, or leptomeningeal metastasis, is a fatal complication of solid tumors. Multimodal analyses of clinical specimens reveal a substantial inflammatory infiltrate enriched for IFN- γ and its downstream signaling. To dissect and overcome this seemingly futile anti-tumor response, we developed syngeneic lung cancer, breast cancer, and melanoma models of leptomeningeal metastasis. Transgenic hosts lacking IFN- γ or its receptor fail to control tumor growth, whereas leptomeningeal overexpression of *Ifng* through a targeted adeno-associated virus system restrains cancer cell growth. Using a suite of transgenic hosts, we show that leptomeningeal T cells generate IFN- γ to recruit and activate peripheral myeloid cells, giving rise to diverse dendritic cell subsets. Independent of antigen presentation, migratory CCR7+ dendritic cells orchestrate the influx, proliferation, and cytotoxic activity of NK cells to control cancer cell growth. While these findings define a leptomeninges-specific IFN- γ circuit, we show that the interferon pathway is preferentially engaged by cDC2 to prime innate immunity, leaving the cDC1 population refractory. We therefore perturbed pathways that selectively license cDC1 maturation, aiming to support the innate, NK cell-centric program with a long-lasting adaptive response.



Supported by SciSerTec

Web: <http://www.scisertec.de>